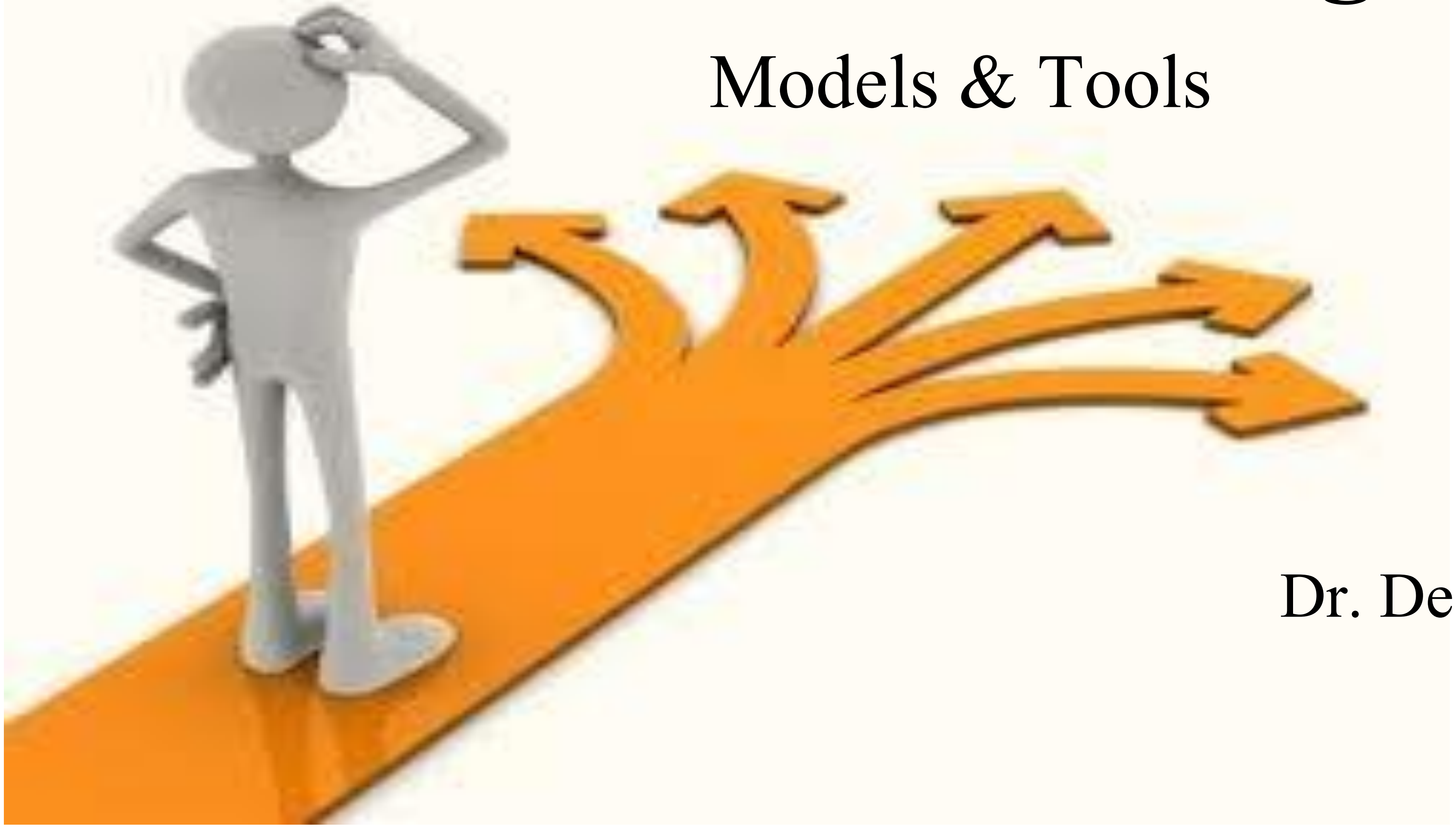
Jnana Prabodhini's
Institute of Psychology
प्रज्ञा – मानस संशोधिका

Sem. V Unit 2.2

Decision Making

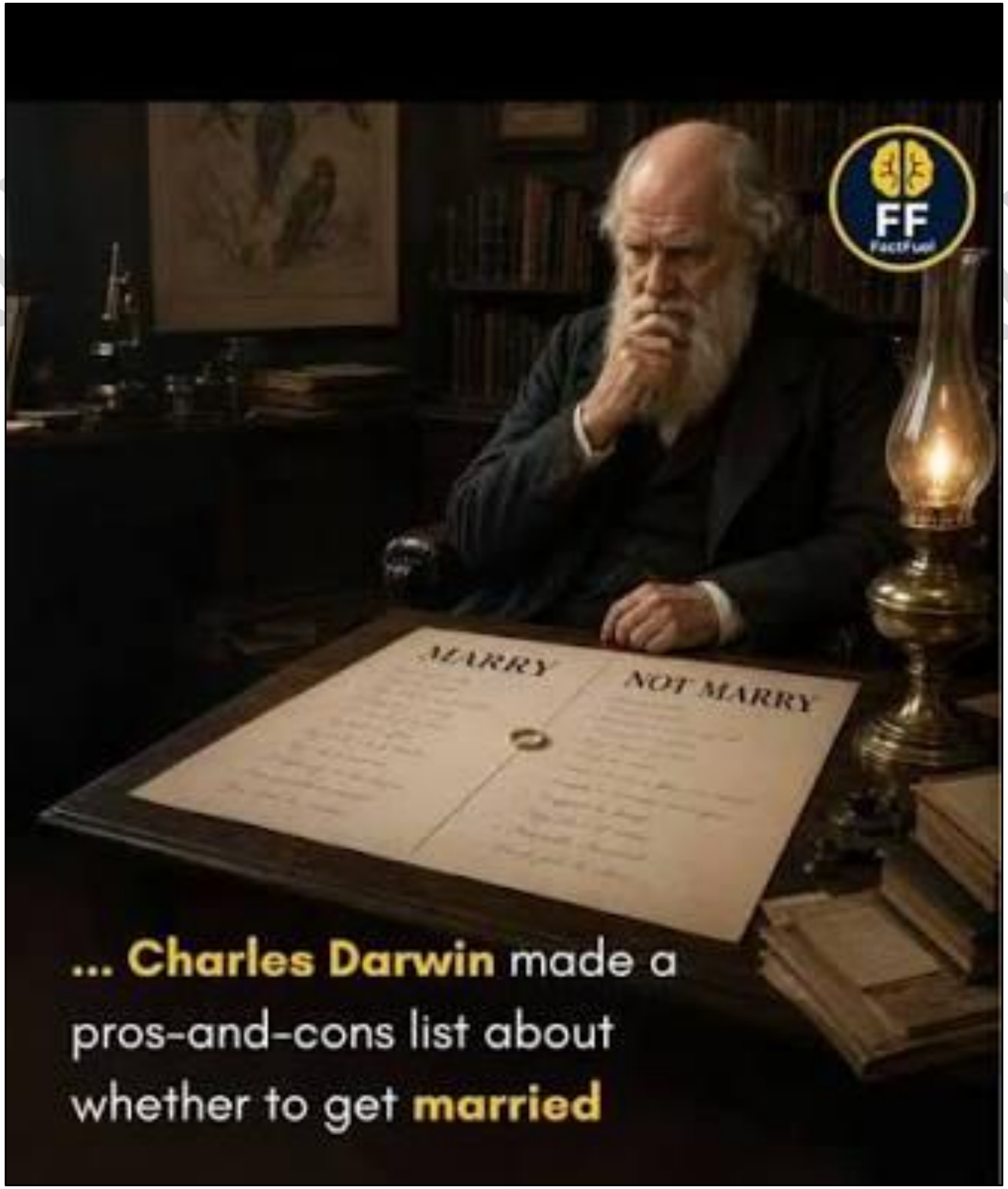
Models & Tools



Dr. Deepak Gupte

Decisions?

- Career?
- College?
- Friends
- Job/Business
- Mobile
- Route to college
- Breakfast


FF
MARRY
NOT MARRY
... Charles Darwin made a pros-and-cons list about whether to get married

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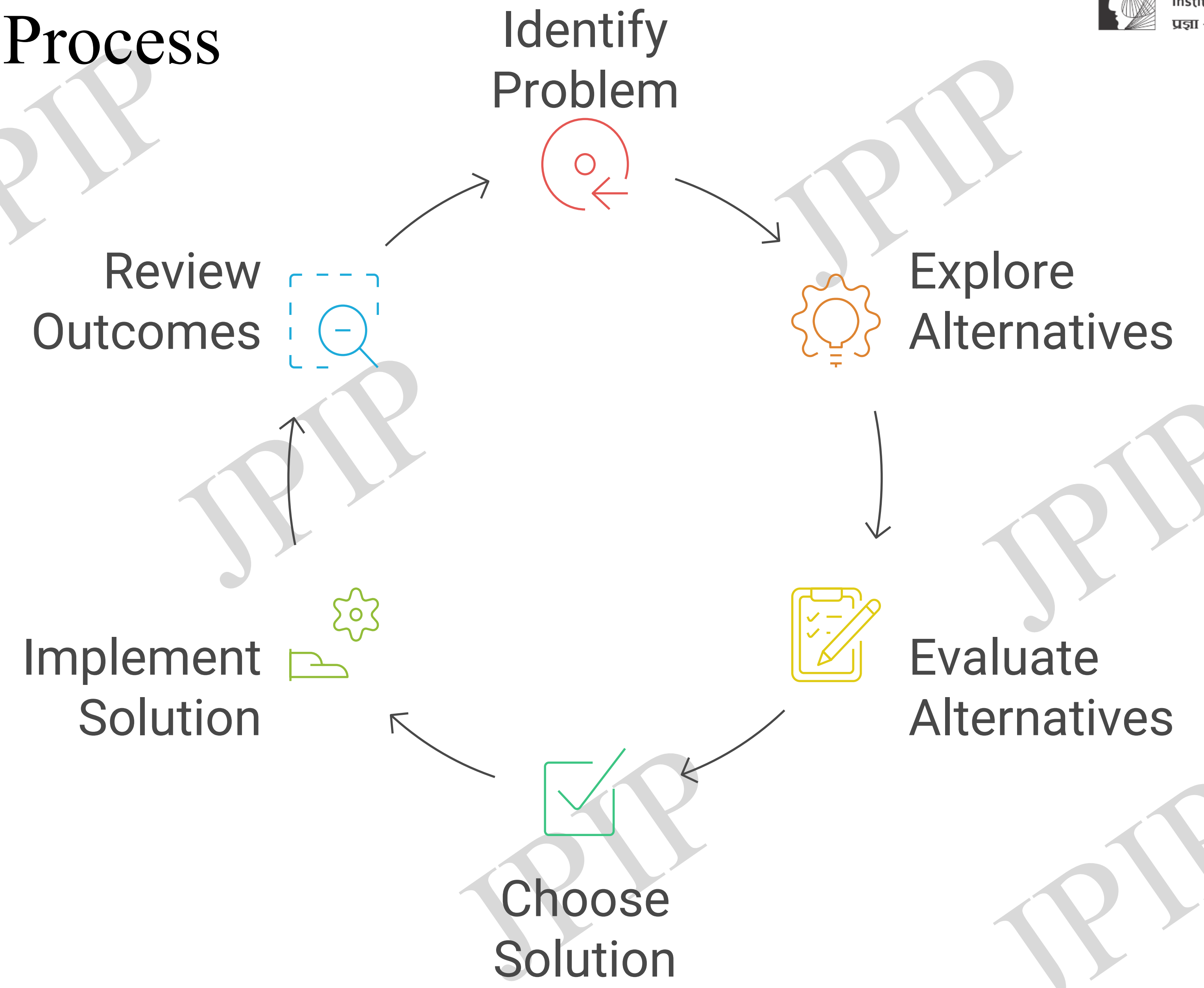
Introduction

- Definition: Selecting from alternatives to solve problems or leverage opportunities (Aamodt, 2013).

- Importance: Drives organizational success, safety, productivity.

Decision Making Process



Schultz & Schultz, 2013

Types of Decision Making Models

1. Rational Model
2. Bounded Rationality
3. Intuitive Model
4. Recognition-Primed model

	Decision-Making Model	Key Words	Pros	Cons	Examples
1	**Rational Model**	**Stepwise**, logical, data-driven, alternatives evaluation	Clear, structured, reduces mistakes	Time-consuming, needs full data	**Civil Engg:** Evaluating multiple bridge designs by analyzing cost, safety, durability, and then choosing the best option. **Mechanical Engg:** Selecting a new CNC machine by comparing efficiency, vendor reliability, and maintenance costs.

	Decision-Making Model	Key Words	Pros	Cons	Examples
2	**Bounded Rationality**	**Information and time are limited** **satisficing**, time pressure, shortcuts	Quick, practical in real-world	May ignore better options, risk of bias	**Electrical Engg:** Choosing a cable supplier quickly when the original supplier is out of stock, based on availability rather than full quality comparison. **IT Engg:** Hiring a programmer under deadline pressure by picking someone "good enough" without fully testing skills.

	Decision-Making Model	Key Words	Pros	Cons	Examples
3	**Intuitive Model**	**Gut feeling**, experience-driven, rapid choice	**Fast**, useful in uncertainty, leverages experience	Subjective, may overlook evidence	**Civil Engg:** A site supervisor deciding to halt concrete pouring due to sudden weather changes, without running full tests. **Mechanical Engg:** An experienced technician sensing from machine vibration that a bearing will fail soon.

	Decision-Making Model	Key Words	Pros	Cons	Examples
4	**Recognition-Primed Model (RPD)**	**Pattern recognition**, expertise, simulation in mind	Fast for experts, effective in emergencies	Requires experience, less effective for novices	**Electrical Engg:** A senior electrician immediately identifying a short circuit cause by recognizing a familiar burning smell and breaker pattern. **IT Engg:** A network engineer quickly diagnosing a server downtime issue by recognizing past failure patterns.

Decision Making tools Decision Matrix

Prioritization Matrix or Weighted Scoring Model

- A structured tool to compare multiple options side-by-side
- Uses pre-defined criteria to evaluate each option
- Criteria are given weights based on importance
- Each option is scored, then total scores help identify the best choice
- Makes the decision process logical, visual, and easy to explain

When to Use a Decision Matrix

- ☑ You have more than one good option
- ☑ Decisions involve multiple, weighted factors (e.g., cost vs. risk vs. speed)
- ☑ You need to compare options fairly and objectively
- ☑ You want to reduce bias and avoid emotional or rushed decisions

Steps to Create a Decision Matrix

1. List the decision options
2. **Identify decision criteria**
3. Assign weights to each criterion
4. Score each option on each criterion
5. **Multiply scores by weights and total**
6. Highest total = best choice

Benefits of the Matrix

- Clear, structured method
- Logically justifies decisions
- Minimizes emotional bias
- Easy to explain to others

Example

Decision: We want to go for a PICNIC.

Criteria: Duration, cost, fun activities, Safety etc..

Options:

☞ Use matrix to decide

Criteria Weightage

- Duration 3
- Cost 4
- Fun activities 5
- Safety 4

Decision Options

Goa

Mulshi

Alibag

Decision Matrix Table Template

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- Fill in scores (e.g., 1 to 5), multiply by weight, and sum across each row.

Options \ Criteria	Criterion 1	Criterion 2	Criterion 3	Criterion 4	**Total Score**
Option A					
Option B					
Option C					
Weights (1–5)					

Choosing a picnic spot

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Options \ Criteria	Duration	Cost	Fun Activities	Safety	Total Score
	Rating x Weightage	Rating x Weightage	Rating x Weightage	Rating x Weightage	
Goa	**2**×**3** = 6	3×4 = 12	5×5 = 25	2×4 = 8	**51**
Mulshi	**5**×**3** = 15	5×4 = 20	3×5 = 15	5×4 = 20	**70**
Alibag	**4**×**3** = 12	4×4 = 16	4×5 = 20	4×4 = 16	**64**

What will you choose? WHY?

Group Activity

- List options & criteria
- Build a matrix as a group
- Discuss results & insights

1. I want to purchase a mobile.
2. I want to choose between job, education, abroad education, start up
3. I want to choose from 3 organizations in campus interview

Decision Making in Teams - Introduction

- Teams pool diverse skills and knowledge.
- Often lead to better quality decisions than individuals.
- Require coordination and open communication.

Advantages & Challenges of Team Decision Making

- Advantages: Multiple perspectives, creativity, shared responsibility.
- Challenges: Groupthink, conflicts, delays.
- Mitigation: Assign roles, encourage dissenting views.

Tools for Team Decision Making

- Brainstorming
- Nominal Group Technique
- Decision Matrices.

1 Nominal Group Technique (NGT)

Structured method for group brainstorming that encourages equal participation and helps generate and prioritize ideas, to minimize the influence of dominant personalities.
Ensuring that everyone's voice is heard during the decision-making process.

1. Silent idea generation:
Participants independently write down their ideas on a specific topic or problem, without any discussion.
2. Round-robin sharing:
Each member shares one idea at a time, which is recorded on a shared surface for everyone to see.
3. Clarification:
Ideas are discussed for clarification and to ensure everyone understands them.
4. Individual ranking:
Each participant privately ranks the ideas based on their preference.
5. Group discussion and decision:
The group discusses the rankings and collectively decides on the most favored ideas.

2 SWOT Analysis - Overview

SWOT Analysis - Practical Exercise

- **Case**: Company launching new product.
- Groups discuss impact on decision making.

Strengths R&D capability	Weaknesses Limited distribution.
Opportunities Growing market	Threats Competitors

SWOT Analysis - Practical Exercise

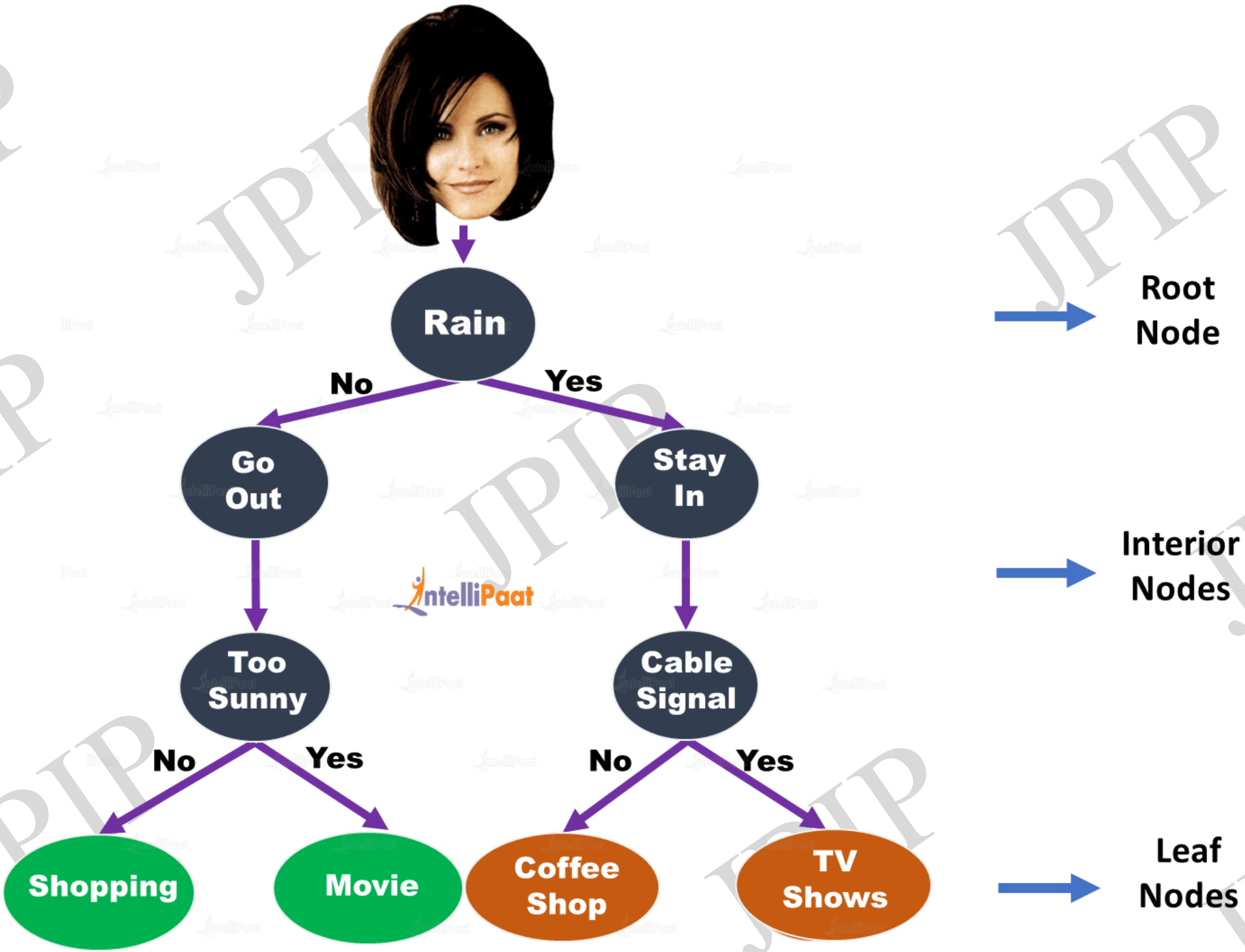
- Whether to participate in theatre competition? College Level
- Hiring new employees as pressure is more in your organization: Org level
- Whether to allow 1 new partner in your startup? Org level

Groups discuss impact on decision making.

Strengths	Weaknesses
Opportunities	Threats

3 Decision Trees - Overview

- Visual representation of decisions, outcomes, and probabilities.
- Helps evaluate risks and rewards.
- Useful for complex choices under uncertainty.



Monica's Weekend Plan

PROJECT MANAGEMENT DECISION HELPER

SHOULD I START THIS PROJECT?

- **YES**
  - Is it important for the company?
    - **YES**
      - Is it too risky?
        - **YES** — Consider not to be involved
        - **NO** — Start the project
    - **NO**
      - Is it important for my career?
        - **YES** — Start the project
        - **NO** — Do not start it
- **NO** — Ask not to be involved

Decision Trees - Practical Example

Supplier choice:

- Supplier A: low cost but uncertain quality
- Supplier B: higher cost but reliable.

- Calculate expected monetary value and make choice.
- Walk through drawing and analysis.

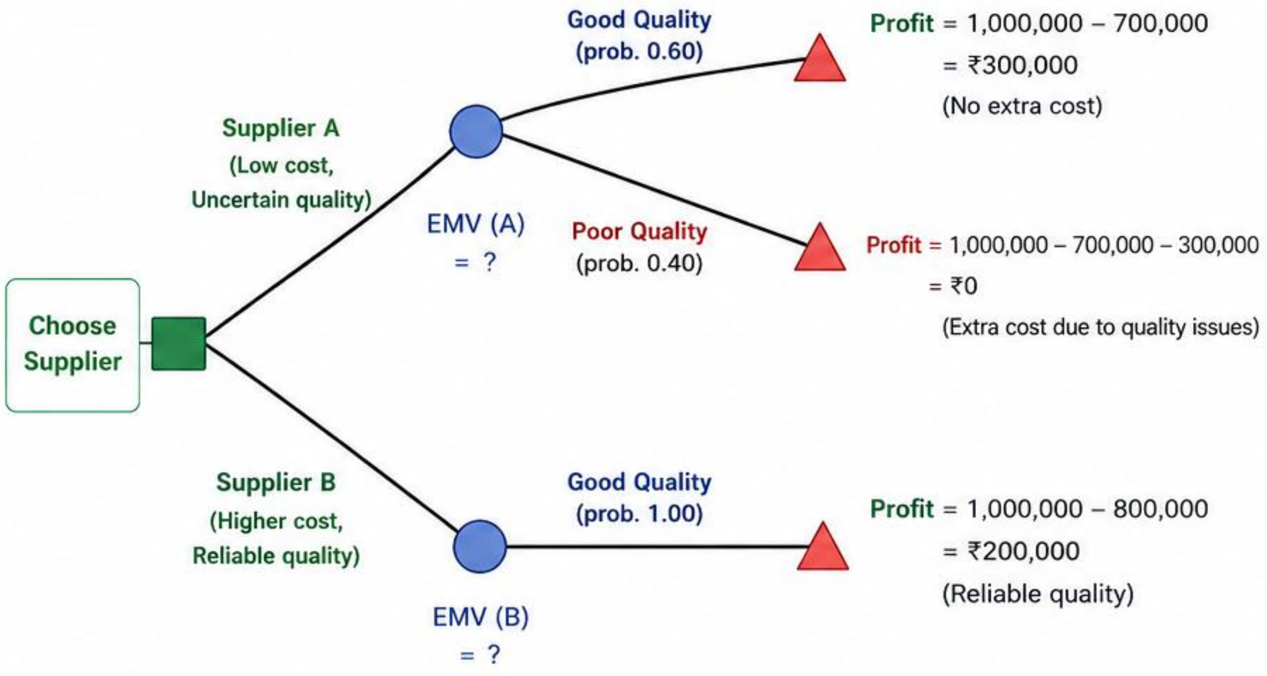
DECISION TREE: SUPPLIER CHOICE

- Supplier A: low cost but uncertain quality
- Supplier B: higher cost but reliable
- Goal: Minimize expected cost (or maximize profit)

Decision Node (□)
Chance Node (○)
Terminal Node (△)

Key Assumptions (Example)

Order quantity value (sales revenue) if all goes well = ₹1,000,000
Cost with Supplier A = ₹700,000
Cost with Supplier B = ₹800,000
If quality is poor (Supplier A):
extra costs due to rework, returns,
delays, penalties = ₹300,000



EXPECTED MONETARY VALUE (EMV) CALCULATION

1) Supplier A

Outcome	Probability	Profit (₹)	Probability × Profit (₹)
Good Quality	0.60	300,000	180,000
Poor Quality	0.40	0	0
EMV (A) = Σ (p × profit) =			**₹180,000**

2) Supplier B

Outcome	Probability	Profit (₹)	Probability × Profit (₹)
Good Quality	1.00	200,000	200,000
EMV (B) = Σ (p × profit) =			**₹200,000**

WALK THROUGH

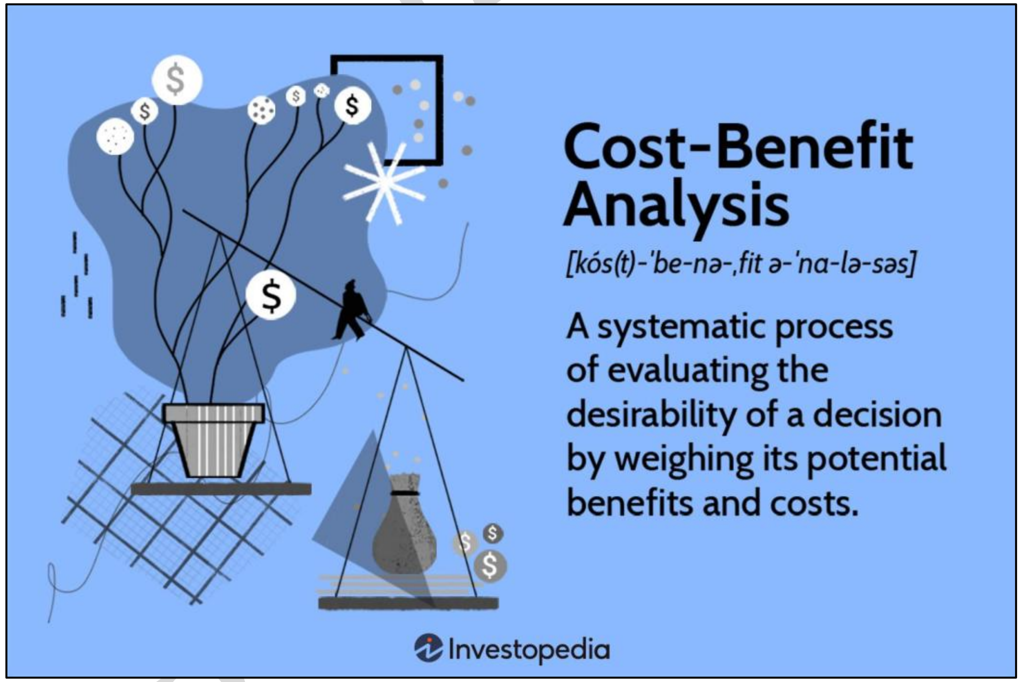
1. Decision Node: We must choose between Supplier A or Supplier B.
2. Chance Nodes: For Supplier A, quality can be Good (60%) or Poor (40%).
   For Supplier B, quality is always Good (100%).
3. Terminal Nodes: Each end point shows the profit (or financial outcome).
4. EMV Calculation: Multiply each outcome by its probability and add.
5. Choose the option with the higher EMV.

DECISION:

Choose Supplier B because its EMV (₹200,000) is higher than Supplier A (₹180,000).

Supplier B is more reliable and gives higher expected profit overall.

Other Decision Making Tools



Other Decision Making Tools

Blue Hat

Successful reflection process

Red Hat

Intuition and feelings

Yellow Hat

Optimistic judgment, Benefits and Hope

White hat

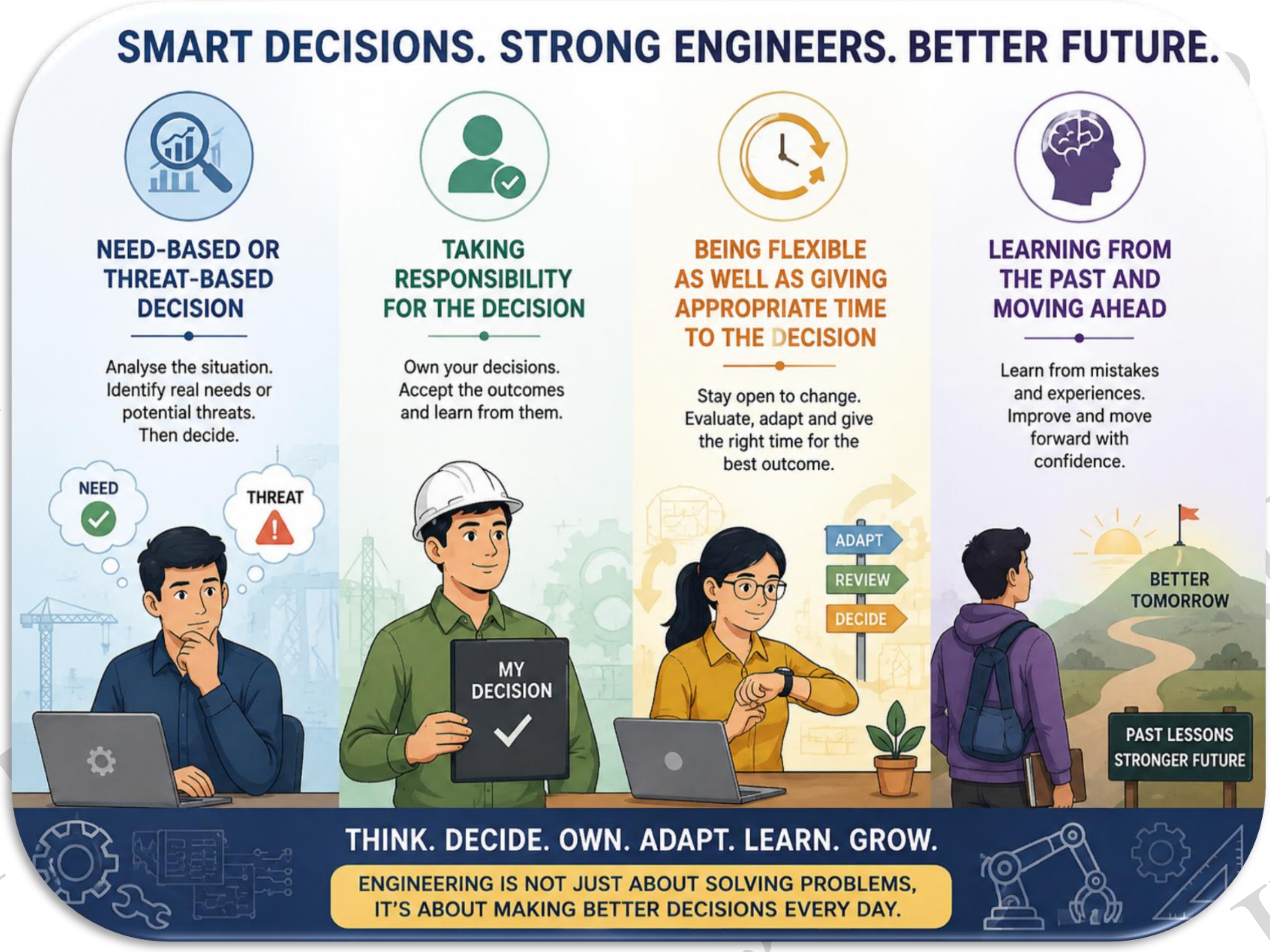
Available and required information

Black Hat

Critical judgement, Risks and Weaknesses

Green Hat

Creative and Alternative Ideas


SMART DECISIONS. STRONG ENGINEERS. BETTER FUTURE.
NEED-BASED OR THREAT-BASED DECISION
Analyse the situation. Identify real needs or potential threats. Then decide.
NEED
THREAT
TAKING RESPONSIBILITY FOR THE DECISION
Own your decisions. Accept the outcomes and learn from them.
MY DECISION
BEING FLEXIBLE AS WELL AS GIVING APPROPRIATE TIME TO THE DECISION
Stay open to change. Evaluate, adapt and give the right time for the best outcome.
ADAPT
REVIEW
DECIDE
LEARNING FROM THE PAST AND MOVING AHEAD
Learn from mistakes and experiences. Improve and move forward with confidence.
BETTER TOMORROW
PAST LESSONS
STRONGER FUTURE
THINK. DECIDE. OWN. ADAPT. LEARN. GROW.
ENGINEERING IS NOT JUST ABOUT SOLVING PROBLEMS, IT'S ABOUT MAKING BETTER DECISIONS EVERY DAY.

Summary of Key Concepts

Decision Making Models

1. Rational Model
2. Bounded Rationality
3. Intuitive Model
4. Recognition-Primed model

Decision Making Tools

1. **Decision Matrix**
2. Decision Tree
3. SWOT
4. Nominal group, PDSA
5. Cost benefit analysis, Six thinking hats

- Review decision models and when to apply.
- Emphasize team decision benefits/tools.
- Importance of practical exercises for mastery.

References and Further Reading

- *Industrial Psychology* Aamodt, M.G. (2013)
- Blink Malcolm Gladwell
- The decisive moment Jonah Lehrer
- Choice by Sheena Iyengar

Application

How will you apply these models and tools in future professional life/internship?